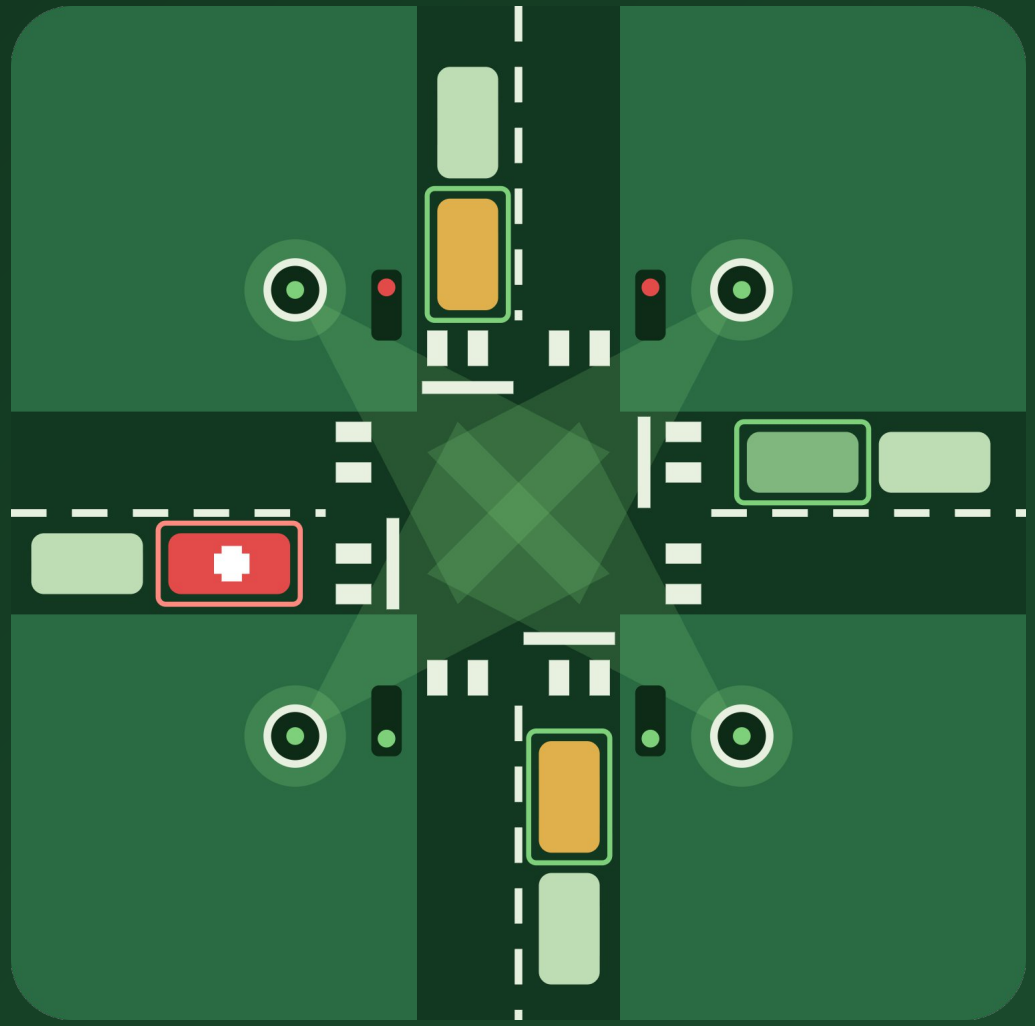
hackFront India Hackathon | Team Name : Quantum Vortex

TrafficSense

An AI-assisted traffic signal system.

Computer Vision | Emergency Priority | Plate Recognition | Arduino Prototype



Four roads · One dashboard · Physical signal model

TrafficSense | TEAM NAME

hackFront India Hackathon | Team Name : Quantum Vortex

Avinaash Bajpai Chhabra

B.Tech Artificial Intelligence & Data Science, Year 2, University School Of Automation & Robotics, GGSIPU EDC

AI | Computer Vision | Database

Yash Kumar

B.Tech Artificial Intelligence & Data Science, Year 2, University School Of Automation & Robotics, GGSIPU EDC

Front End | 2nd C.V (Help) | Dashboard

Ridhima Tiwari

B.Tech Artificial Intelligence & Data Science, Year 2, University School Of Automation & Robotics, GGSIPU EDC

Traffic Logic | Arduino | Integration

Pranay Nigam

B.Tech Artificial Intelligence & Data Science, Year 2, University School Of Automation & Robotics, GGSIPU EDC

Backend

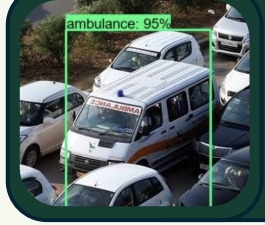
THE PROBLEM

In India, traffic jams, red lights, and blocked roads can delay ambulances and other emergency vehicles, potentially leading to preventable deaths and permanent disabilities.

This includes:

- Ambulances: Patients dying before reaching hospitals.
- Fire engines: Delayed arrival allowing fires to spread, causing deaths, burns, and property damage.
- Police and rescue vehicles: Delayed response to violent incidents or emergencies.
- Organ transport vehicle: A donated organ is delayed in heavy traffic while being transported to a transplant hospital causing risk Risk of organ damage or transplant cancellation
- Pregnant women: Delays in cars, auto-rickshaws, or ambulances leading to dangerous childbirth complications.

A 2026 study of 626,372 ambulance hotline calls in Punjab and Haryana during 2025 found that 33% of patient transfers took more than 60 minutes from the initial call to hospital arrival.



A 45yr old man, suffered a heart attack. His ambulance moved only around 100 metres in 30 minutes because of congestion. He died before reaching hospital.
Delhi, India (2017) [Source : Times Of India]



A 32-year-old woman died of a heart attack in Kanpur, India, after traffic congestion delayed her arrival at the hospital by 45 minutes.
Kanpur, India (2025) [Kanpur News]



Three died in a Puppalguda fire; a responding fire engine faced a five-minute traffic delay, but the report doesn't link this delay to the deaths.
Hyderabad, India (2025) [Times Of India]

Fig1 - News showing the Problem

SOLUTION

Traffic congestion and fixed traffic signals delay emergency vehicles when every second matters. Intersections need smarter signal control to prioritize emergency passage and reduce critical delays.

TrafficSense : An AI-assisted traffic signal system

An AI-assisted traffic management system that turns live camera feeds into smarter signal recommendations.

- Adaptive signals: Suggests green-light durations based on traffic density.
- Emergency priority: Prioritizes ambulances, fire engines and police vehicles.
- Violation monitoring: Flags possible violations with vehicle images, timestamps and readable number plates.
- Unified dashboard: Brings live monitoring, signal recommendations and incident review into one place.

**Using this, we are also solving the problem of unnecessary waiting at red lights when other roads at the intersection are empty.*

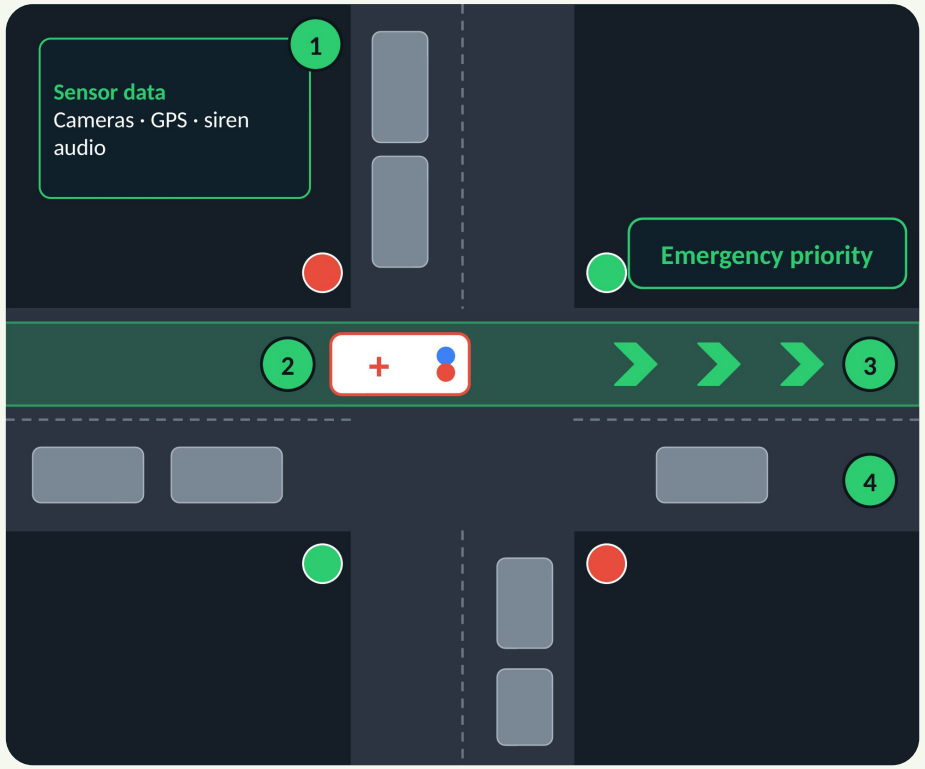
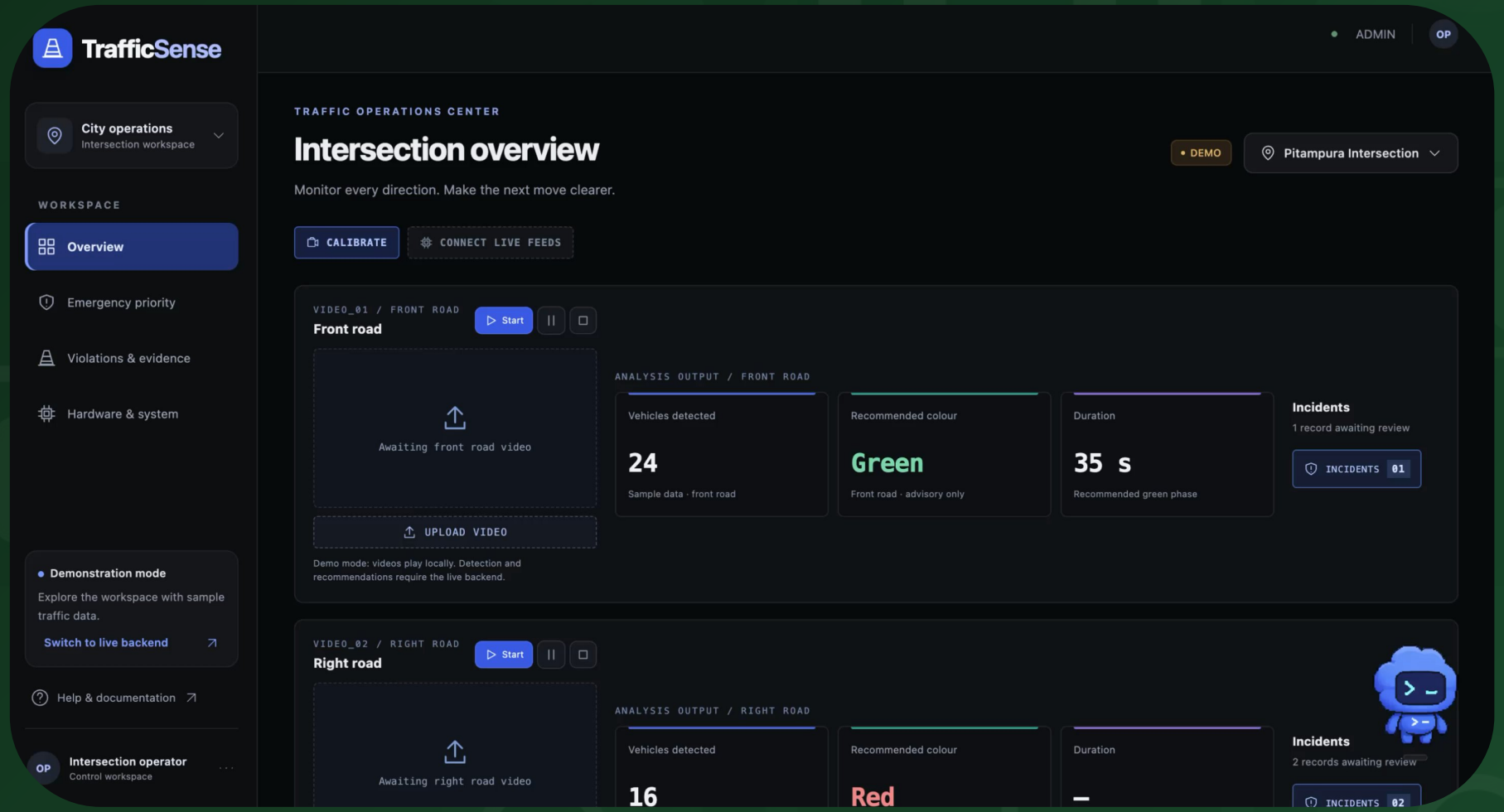


Fig2 – Depiction of the Solutin

TrafficSense | DASHBOARD

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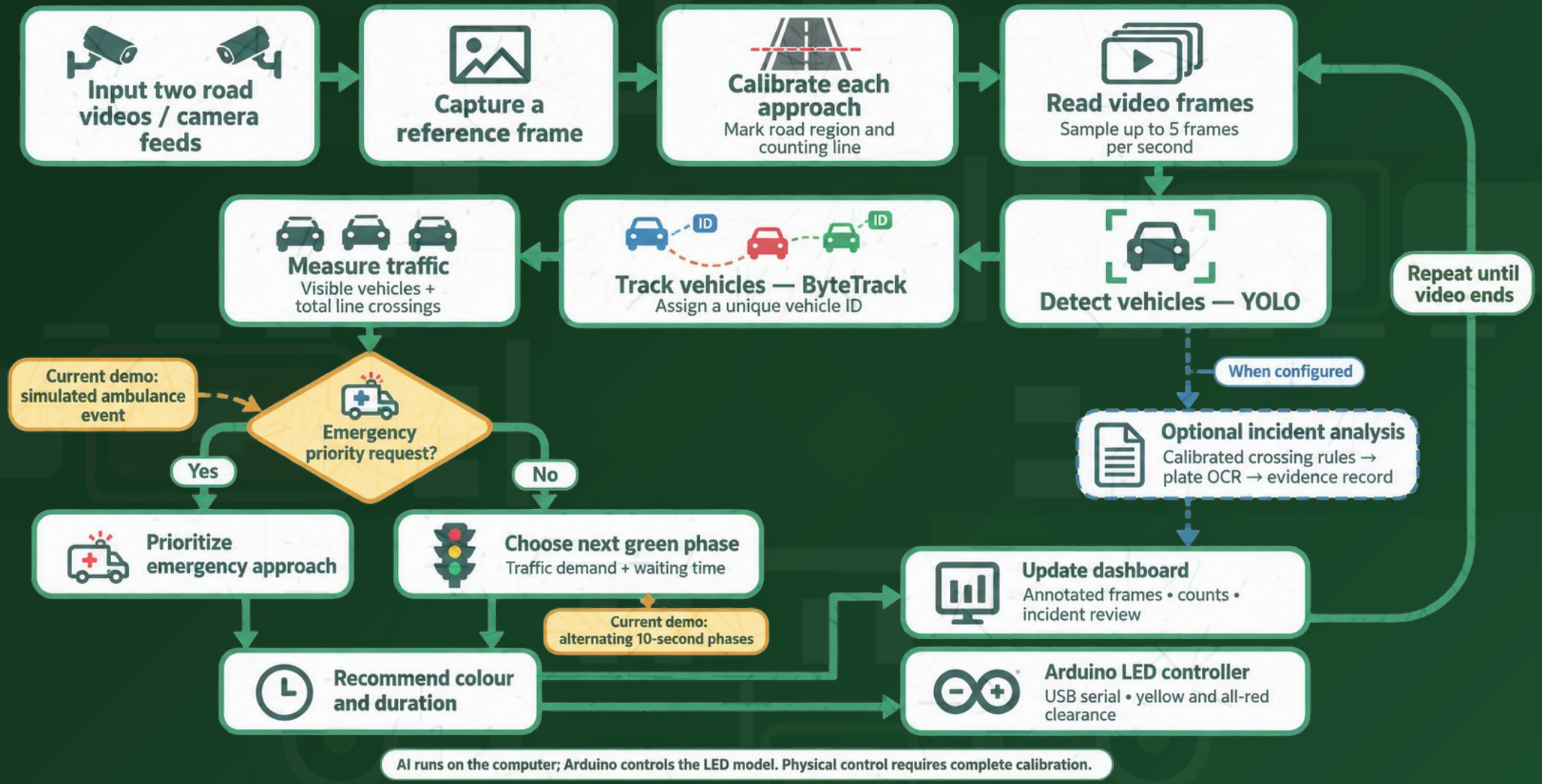
Here in the Right Road 10sec is pinging back-to-back as the demo video is over.

**Arduino Live has not been used it is not possible to connect to live red light and vehicles at the moment.

Once the Arduino is connected it LIVE Video picking, calibration & other features will get activated.

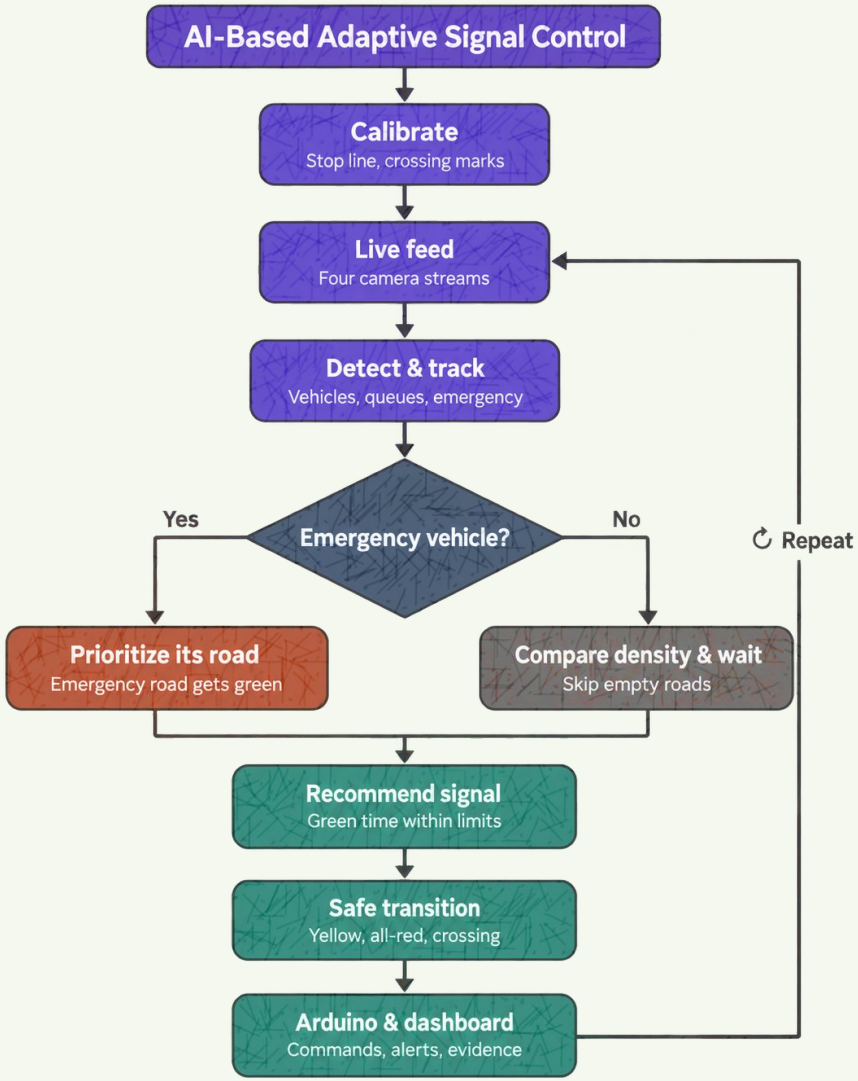
TrafficSense | EXPLANATION

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AI-Based Adaptive Signal Control

1. **Calibrate:** Capture a frame from each camera and mark the stop line, traffic line and pedestrian crossing.
2. **Detect & track:** Analyse all four live feeds to identify vehicles, waiting traffic and emergency vehicles.
3. **Select priority:** If an emergency vehicle is detected, prioritize its road. Otherwise, choose based on vehicle density and waiting time, skipping empty roads.
4. **Recommend timing:** Suggest a green-light duration proportional to traffic demand, within minimum and maximum limits.
5. **Switch safely:** Pass through yellow and all-red clearance before giving another road green. Hold conflicting traffic while pedestrians clear the crossing.
6. **Monitor & repeat:** Update the dashboard, send signal commands to Arduino and record possible violations with image evidence and readable plates.



FEATURES

AI-Based Adaptive Signal Control

01

Calibrate Once

Mark road boundaries directly on a camera frame.

02

Smarter Signals

Recommend green time based on traffic demand.

03

Emergency First

Prioritize detected emergency vehicles.

04

Violation Evidence

Capture possible violations with photos and plate details.

05

One Dashboard

Monitor four roads and review incidents together.

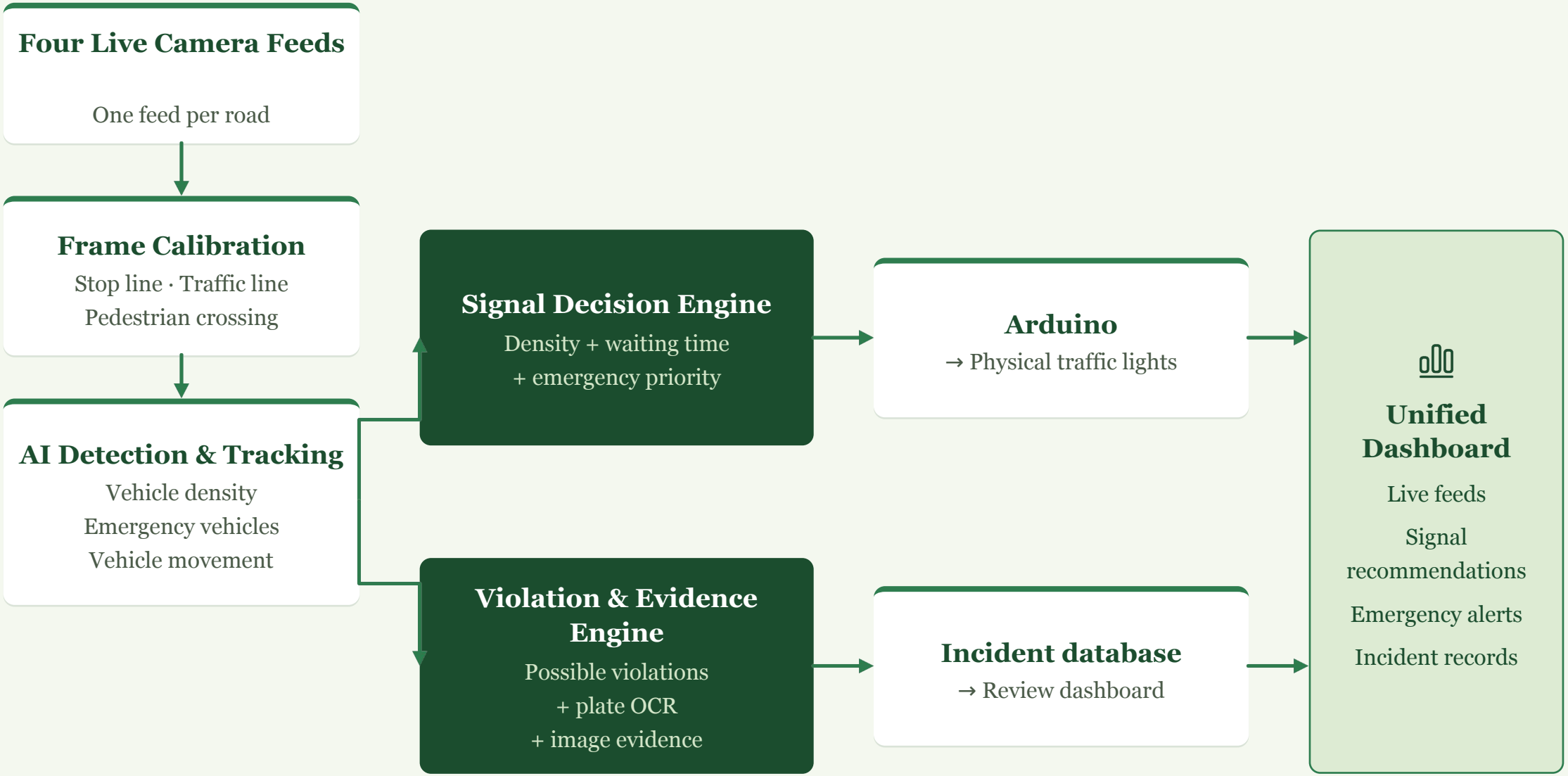
06

Arduino Integration

Demonstrate signal decisions through physical traffic lights.

Camera → AI detection → Signal decision → Arduino & dashboard

SYSTEM ARCHITECTURE



The laptop performs AI analysis; Arduino controls the physical signal model.

TECHNOLOGY STACK

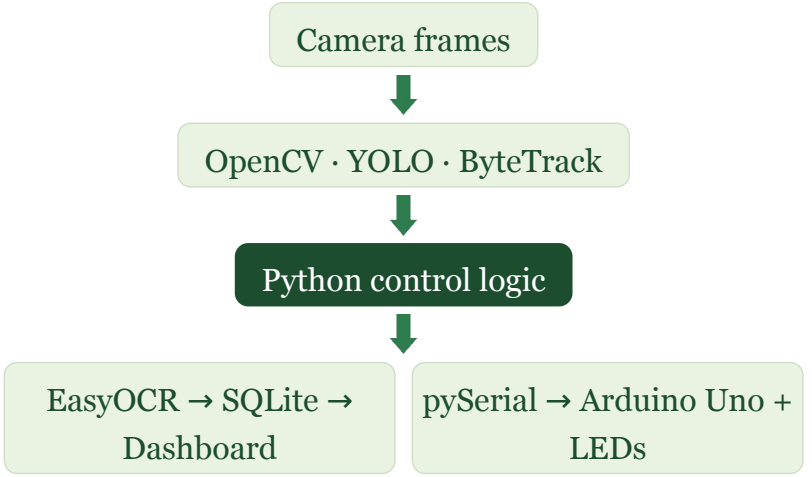
Layer	Technology
Dashboard	HTML, CSS, JavaScript
Backend & Control Logic	Python
Computer Vision	OpenCV, YOLO, ByteTrack
Number Plate Recognition	EasyOCR
Database	SQLite
Hardware	Arduino Uno, LEDs, embedded C++
Communication	USB Serial using pySerial

7
layers, from dashboard to hardware

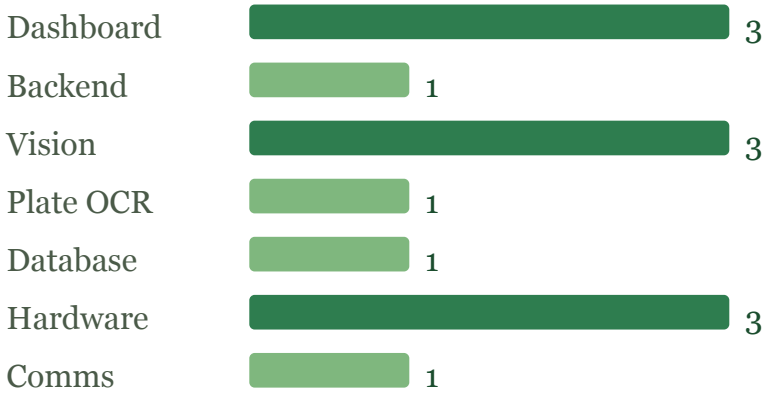
Local
analysis runs on the laptop

USB
serial link to the Arduino

How the stack fits together

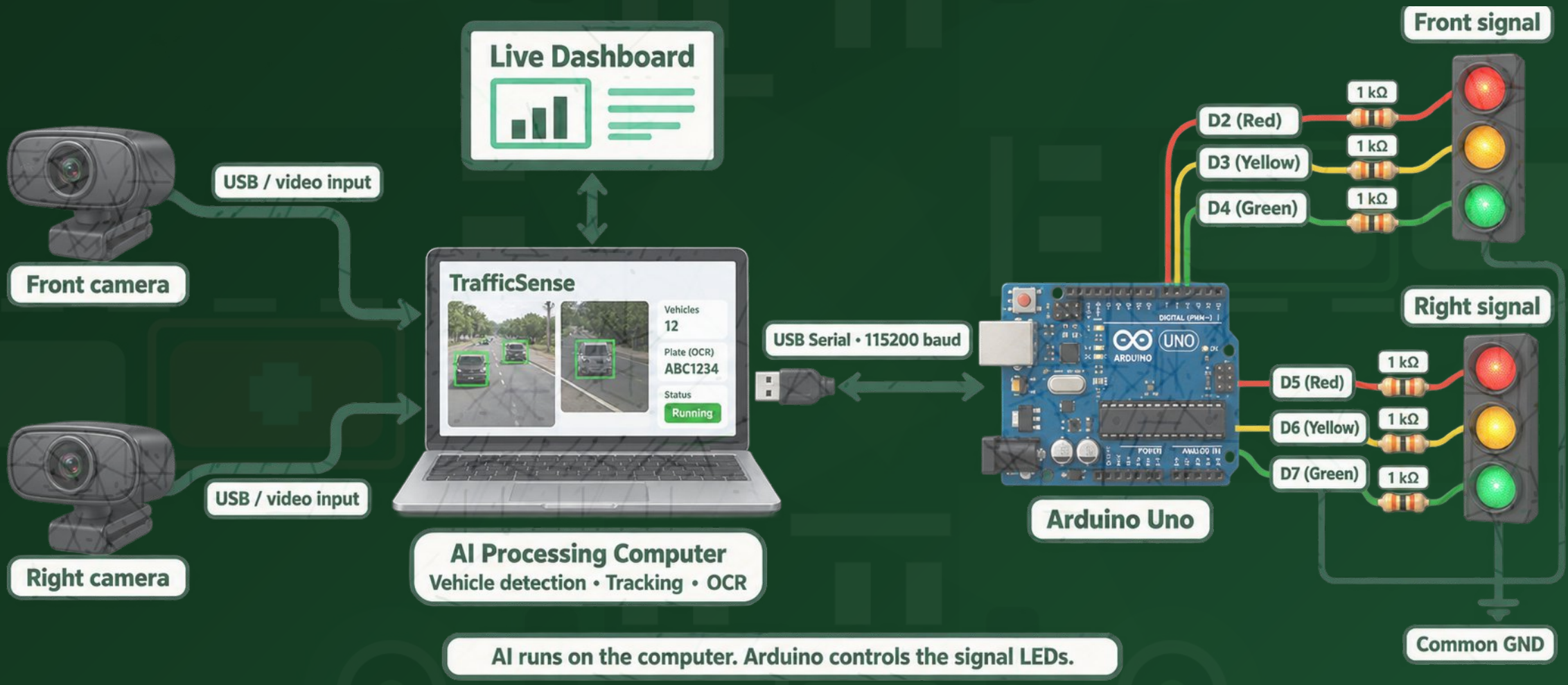


Technologies listed per layer



TrafficSense | SCHEMATICS [ARDUINO HARDWARE]

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DATABASE-ER DESIGN

Current prototype: one SQLite “Violations” table

Group	What it stores
Identity	Event ID, session ID, road, vehicle tracking ID
Incident	Violation type, location, timestamps
Evidence	Full-frame image, vehicle crop, plate crop
Recognition	Plate text and confidence scores
Review	Needs review, confirmed or dismissed

What each record stores

Items listed per group



Proposed Database Design *Not separate tables in the current implementation*

Intersection
Intersection ID
Name and location

→

Cameras
Camera ID
Road
Calibration lines

→

Analysis Sessions
Session ID
Start and end time
Camera feeds

→

Incident Records
Event ID · Tracking ID
Violation type
Timestamps
Review status

→

Evidence
Full-frame image
Vehicle crop
Plate crop
Plate text and scores

Note: images are stored as files, with their paths in SQLite.

FEASABILITY & VIABILITY

Practical prototype
Runs on a laptop with camera inputs and an Arduino signal model.

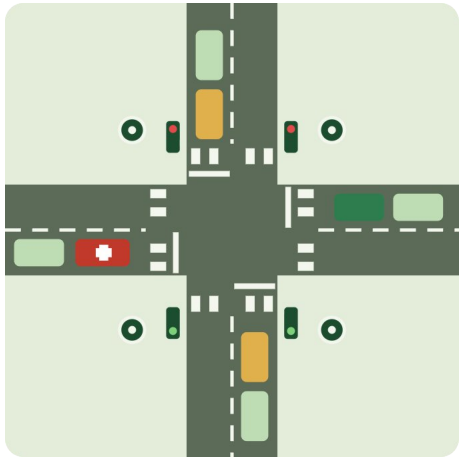
Affordable demonstration
Uses open-source software and readily available electronic components.

Adaptable setup
Frame calibration configures the system for different fixed-camera intersections.

Local processing
Core analysis can operate without continuous internet access.

Deployment requirements
Real-world use needs validated models, sufficient computing power and integration with approved signal hardware.

Four fixed cameras, one intersection



Prototype vs real-world use

	Prototype	Deployment
Compute	Laptop	Sufficient computing power
Models	Open-source	Validated models
Signals	Arduino model	Approved hardware

IMPACT & BENEFITS

Faster emergency passage
Through priority signal recommendations.

Less unnecessary waiting
When competing roads are empty.

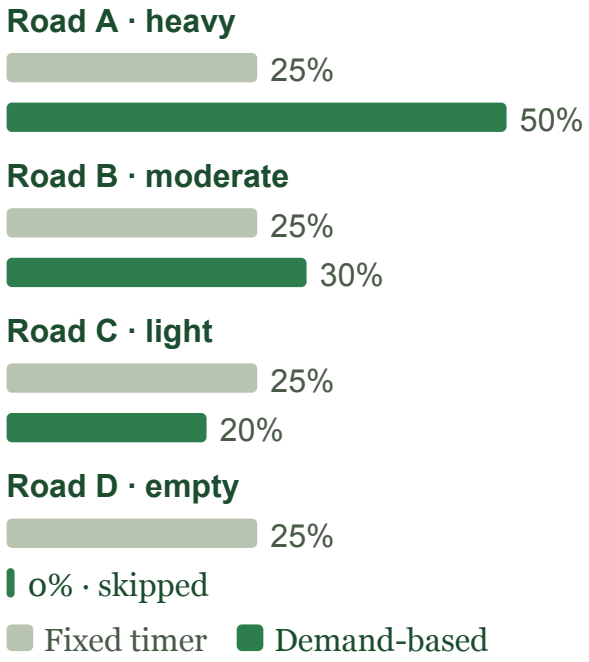
More responsive signals
Green time follows traffic demand.

Better incident review
Timestamped images and plate details.

Greater traffic visibility
One dashboard for four roads.

Fixed timer vs demand-based green time

Illustrative concept with example demand 5 : 3 : 2 : 0, not measured data

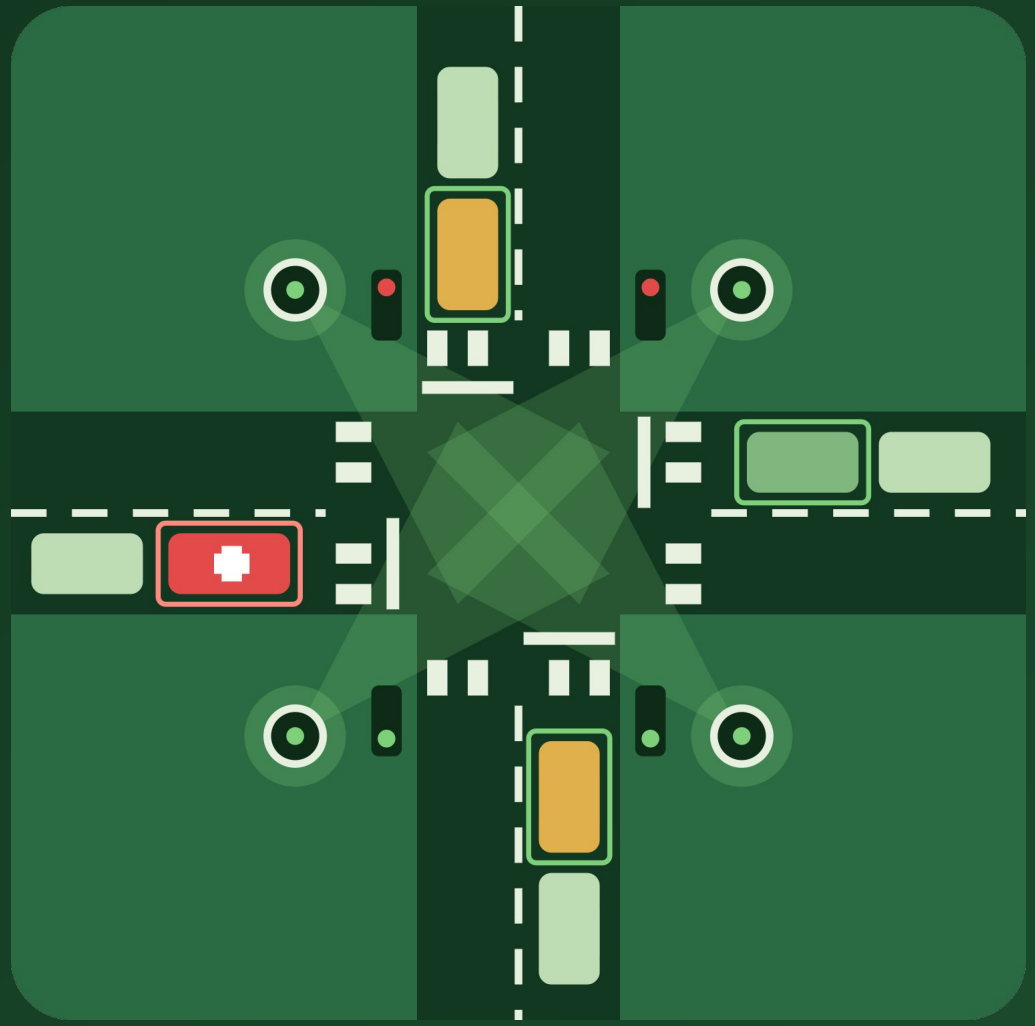


These benefits require field evaluation; traffic-delay reductions have not yet been measured.

TrafficSense

THANK YOU

QUANTUM VORTEX



Four roads · One dashboard · Physical signal model